Munwar Mohammad

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Professional Summary

AI/ML Engineer with 4+ years of experience progressing from fraud detection and anomaly modeling at Wells Fargo to production NLP, forecasting, and multi-agent Generative AI systems at FedEx. Builds end-to-end AI systems on Microsoft Azure: PySpark data pipelines, PyTorch and XGBoost models, agentic RAG with LangGraph, evaluation-gated releases in MLflow, and monitored inference services on AKS. Reduced operational information lookup time by 62% with a multi-agent RAG assistant serving 280 operational guides.

Skills

Generative AI and Agentic Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, Azure OpenAI, OpenAI API, Prompt Engineering, Embeddings, Hybrid Search, Reranking, Tool Calling, Structured Outputs

LLM Evaluation and Safety: LLM Evaluation, LLM-as-Judge, LLM Guardrails, Prompt Injection Filtering, PII Redaction, Human-in-the-Loop Review, Langfuse

Deep Learning, NLP and Machine Learning: PyTorch, TensorFlow, Deep Learning, Natural Language Processing (NLP), Transformers, Hugging Face, Scikit-learn, XGBoost, Classification, Regression, Imbalanced Learning (SMOTE), Anomaly Detection, Isolation Forest, Time-Series Forecasting, Prophet, Feature Engineering, Cross-Validation, Hyperparameter Tuning, SHAP

Programming and Data Engineering: Python, SQL, Pandas, NumPy, PySpark, Azure Databricks, Azure Data Factory, Azure Data Lake Storage Gen2 (ADLS Gen2), Delta Lake, Databricks Vector Search, Data Quality Validation

Cloud and MLOps Engineering: Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP), Amazon SageMaker, Amazon Bedrock, Google Vertex AI, MLflow, FastAPI, Docker, Kubernetes (AKS), Azure ML Endpoints, Model Monitoring, Data Drift Detection, Evidently AI, Application Insights, Azure DevOps CI/CD, PyTest, Git, PostgreSQL, Redis

Work Experience

AI/ML Engineer Jun 2025 - Present

FedEx *US*

- Engineered a multi-agent RAG assistant using LangGraph, Azure OpenAI, and Databricks Vector Search, with dedicated agents for SOP retrieval, exception analysis, and answer validation across 280 operational guides; hybrid retrieval and reranking reduced average information lookup time from 8.5 to 3.2 minutes (62%).
- Built an NLP exception-triage model using PyTorch and Hugging Face Transformers to classify unstructured driver notes into 12 root-cause categories, combining predictions with shipment scan-history data and improving classification recall by 9% over the previous workflow.
- Created a 1,200-example LLM evaluation dataset measuring retrieval relevance, groundedness, citation accuracy, and classification precision; calibrated LLM-as-judge scores against human labels and enforced MLflow evaluation results as release gates for every prompt and model change.
- Built PySpark batch-scoring pipelines on Azure Databricks, ADLS Gen2, and Azure Data Factory to score daily shipment records across three hubs and prioritize packages at risk of delivery exceptions.
- Developed Prophet and XGBoost forecasting models for shipment volumes across six regional hubs, using backtesting to reduce MAPE from 18.6% to 13.2% and improve capacity planning against demand shifts.
- Deployed LLM and ML inference services with FastAPI and Docker through Azure ML endpoints on Kubernetes (AKS), implementing versioned prompts, PII redaction, prompt-injection filtering, and schema validation for safe, repeatable releases.
- Implemented model and service monitoring with Evidently AI and Application Insights covering data drift, latency, token consumption, and errors, reducing incident triage time from 4 hours to under 3.
- Automated PyTest-based testing and deployment checks in Azure DevOps CI/CD pipelines for model and application releases.

Machine Learning Engineer Jul 2021 - Jun 2024

Wells Fargo *IN*

- Developed an XGBoost fraud-classification model on imbalanced transaction data using SMOTE, class weighting, and threshold optimization, improving F1-score by 7% over the baseline model.
- Implemented an Isolation Forest anomaly-detection workflow to prioritize approximately 1,250 monthly alerts; investigator feedback showed 31% of flagged alerts warranted further action.
- Built reusable feature-engineering workflows with Python, SQL, Pandas, and NumPy, adding schema checks and missing-value handling that reduced manual data-preparation effort by 20% across monthly model-refresh cycles.
- Evaluated candidate models using Scikit-learn, stratified cross-validation, hyperparameter tuning, and SHAP analysis to examine segment-level errors and feature influence; findings drove threshold adjustments and clearer model explanations for risk and model-validation stakeholders.

Projects

Multi-Agent GitHub Code Reviewer

- Built a LangGraph multi-agent code-review system in which a supervisor routes pull requests to static-analysis, security, architecture, and code-quality agents, consolidating outputs through the OpenAI API into deduplicated, structured pull-request feedback.
- Implemented asynchronous GitHub webhook processing with FastAPI, Celery, Redis, and PostgreSQL in Docker; added Langfuse tracing for agent execution and token consumption, plus Prometheus and Grafana dashboards for job throughput and latency.

Agentic RAG Platform for SEC Filings

- Built an agentic RAG platform over SEC 10-K filings from 20 companies, using LangGraph for agent orchestration with LangChain and LlamaIndex for hybrid retrieval, reranking, tool calling, conversation memory, and structured outputs, routing low-confidence answers to human review.
- Implemented document ingestion, chunking, and embeddings on Azure Databricks with Delta Lake tables governed in Unity Catalog and indexed in Databricks Vector Search; routed model calls through Mosaic AI Gateway with guardrails and citation validation, evaluated retrieval relevance and groundedness, and automated deployment with GitHub Actions.

Education

University of Alabama at Birmingham

MS, Data Science *May 2026*